

Chapter 12: Securing Networks & Wireless Systems

CompTIA Security+ Study Slides

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Learning Objectives

- Understand **wired vs. wireless network vulnerabilities**
- Explain **WEP, WPA, WPA2, and WPA3** security protocols
- Recognize **WPS weaknesses** and disable insecure features
- Identify **rogue access points** and **evil twin attacks**
- Apply wireless hardening and monitoring best practices

Overview

Wireless networks introduce unique security challenges:

- Signals travel through the air (no physical boundary)
- Devices join dynamically
- Users expect mobility and convenience

 **Goal:** balance accessibility with strong encryption and authentication.

Wired vs. Wireless Security

Aspect	Wired Network	Wireless Network
Medium	Copper/fiber cables	Radio frequency (RF)
Access Control	Physical port security	Open air; harder to restrict
Threat Surface	Local access only	Extended range (up to 300m+)
Primary Risks	Sniffing, port scanning	Eavesdropping, rogue APs, jamming

 *Wireless attacks can originate from outside physical premises.*

Wi-Fi Security Evolution

Protocol	Key Features	Weakness
WEP (Wired Equivalent Privacy)	RC4 stream cipher, 24-bit IV	Easily cracked within minutes
WPA (Wi-Fi Protected Access)	TKIP, message integrity (MIC)	Legacy, weak by today's standards
WPA2	AES-CCMP encryption, 802.11i standard	Strong, but vulnerable to KRACK
WPA3	SAE handshake, forward secrecy	Current standard; resists offline brute-force

 Always use **WPA3 (or WPA2-AES)** — never use WEP or TKIP.

WEP — Wired Equivalent Privacy

- Introduced in 1999 (IEEE 802.11b)
- Uses **RC4** encryption and a **24-bit Initialization Vector (IV)**
- Predictable IVs → repeated keys → crackable with Aircrack-ng

Why WEP Failed:

- Short IV reuse leads to key collisions
- Weak checksum (CRC-32)
- Authentication bypass possible

 **Deprecated:** should be completely disabled.

WPA and WPA2

WPA (2003)

- Introduced TKIP (Temporal Key Integrity Protocol)
- Included MIC for message integrity
- Designed as a quick fix for WEP vulnerabilities

WPA2 (2004)

- Replaced TKIP with **AES-CCMP encryption**
- Required by Wi-Fi Alliance since 2006
- Still widely used today

 WPA2-Enterprise supports 802.1X with RADIUS authentication.

WPA3 Enhancements

- Uses **Simultaneous Authentication of Equals (SAE)** instead of PSK
- Provides **Forward Secrecy** — session keys unique per connection
- Protects against **offline dictionary attacks**
- Introduces **192-bit encryption mode** for enterprise environments

 WPA3 = stronger authentication + improved resilience to brute-force attacks.

WPA Personal vs. Enterprise

Mode	Description	Authentication
WPA/WPA2-Personal (PSK)	Pre-shared key shared among users	Password-based
WPA/WPA2-Enterprise (802.1X)	Centralized authentication via RADIUS server	Username/password or certificate

 Enterprise mode allows individual user revocation — critical for corporate Wi-Fi.

Wi-Fi Protected Setup (WPS)

WPS simplifies device connection with an 8-digit PIN or push-button setup.

Vulnerability:

- The 8-digit PIN can be brute-forced in hours.
- Many routers fail to lock out repeated attempts.
- Exposes network even with WPA2 enabled.

Best Practice:

 **Disable WPS** entirely on all access points.

Rogue Access Points (Rogue AP)

Definition: unauthorized AP connected to the network — either malicious or accidental.

Risks:

- Bypasses firewall and NAC
- Enables sniffing and credential theft
- May connect to attacker's laptop acting as gateway

Detection:

- Wireless Intrusion Detection System (WIDS)
- Regular site surveys (e.g., AirMagnet, Kismet)

Evil Twin Attack

Evil Twin: a rogue AP mimicking a legitimate SSID to trick users into connecting.

Steps:

1. Attacker sets up fake AP with same SSID.
2. Victim connects automatically.
3. Attacker captures credentials or injects malware.

Prevention:

- Use **VPN** for all Wi-Fi sessions
- Validate certificate of authentication server (RADIUS)
- Educate users not to connect to open networks

Wireless Jamming

Definition: deliberate transmission of radio interference to disrupt Wi-Fi communication.

Methods:

- Continuous RF noise generation
- Deauthentication floods
- Targeted channel interference

Defense:

- Spectrum analysis to locate jammer
- Switch channels or frequencies
- Use **5 GHz** or **6 GHz Wi-Fi** bands for resilience

Initialization Vector (IV) Attacks

- Exploits repeated IVs in WEP and early WPA
- Enables attackers to recover encryption keys
- Tools: Aircrack-ng, Kismet

Countermeasure:

- Use **WPA2-AES** or **WPA3-SAE**
- Rotate keys frequently
- Disable legacy protocols on APs

Deauthentication & Disassociation Attacks

Attackers send forged **deauth frames** to disconnect users from Wi-Fi.

Purpose:

- Force reconnect to **evil twin**
- Capture 4-way handshake for password cracking

Mitigation:

- WPA3's **Protected Management Frames (PMF)**
- Monitor for excessive deauth requests
- Use WIDS/WIPS for real-time detection

Wireless Access Point (WAP) Security Checklist

- ✓ Change default SSID and admin password
- ✓ Disable WPS and UPnP
- ✓ Use WPA2-AES or WPA3 encryption
- ✓ Enable MAC filtering and DHCP reservations
- ✓ Restrict signal range (lower transmit power)
- ✓ Keep AP firmware updated
- ✓ Log and monitor wireless connections



Wireless Security Monitoring

- Deploy **Wireless IDS/IPS** to detect anomalies
- Log all connections and signal strengths
- Integrate with **SIEM** for correlation
- Watch for:
 - Unknown SSIDs
 - High failed authentication rates
 - Sudden signal spikes (potential jamming)

Wireless Network Segmentation

Separate wireless traffic based on trust and purpose:

- **Corporate WLAN:** secured, authenticated users
- **Guest Network:** Internet-only, VLAN-isolated
- **IoT Network:** limited access, restricted routing
- **Management VLAN:** AP configuration only

 *Segmentation prevents lateral movement and containment breaches.*

Wi-Fi Hardening Summary

Category	Best Practice
Encryption	Use WPA3 or WPA2-AES
Authentication	Implement 802.1X / RADIUS
Access Control	Disable WPS, enforce strong passphrases
Monitoring	WIDS/WIPS, SIEM integration
User Awareness	Train users on SSID spoofing and public Wi-Fi risks

Key Takeaways

- **Disable outdated protocols** (WEP, WPA-TKIP, WPS)
- **Use WPA3 with AES and Protected Management Frames**
- **Detect rogue APs and evil twins** with WIDS/WIPS
- **Segment wireless networks** by purpose
- Continuous monitoring and firmware patching maintain resilience

End of Chapter 12

Next: [Chapter 13 — Physical & Facility Security](#)